Different Parametric Updates

A) SGD (Stochastic Gradient Descend)

B) SGD + Momentum (momentum update)

C) Nesterov momentum update

D) Adam

E) Adam W

F) RmsProp

G) Adagrad

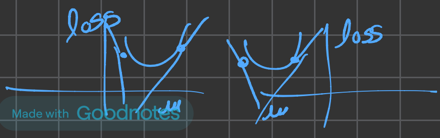
H) Adadelta

I) Muon **

Parametric update?

(A) forward
(B) loss
(C) zero grad
(D) Backward Pass
(E) Parameter updates

Assume loss is +ve



$$\boxed{w_t = w_t - (\alpha * dw_t)}$$

$\rightarrow$ SGD $\Rightarrow -(\alpha * dw_t)$

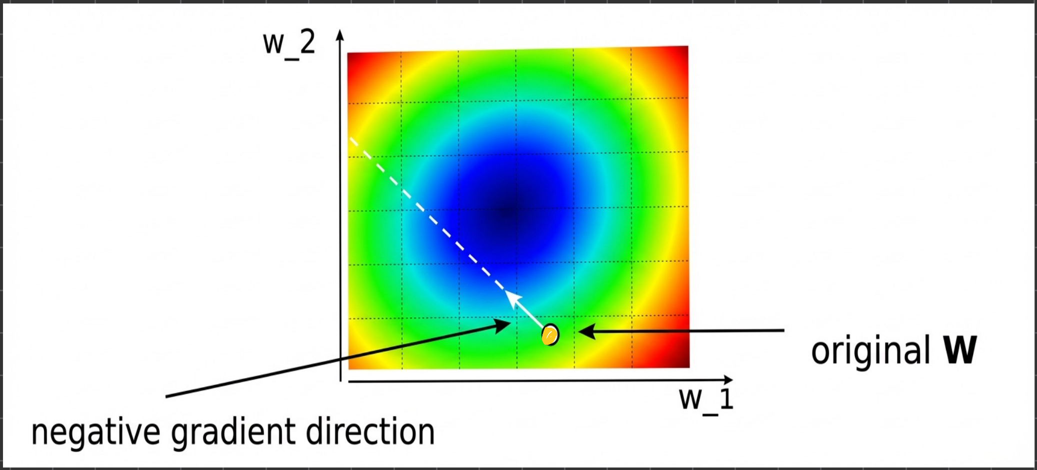
α = learning rate

A) headache (It is an art + science to get it working)

B) one of the most critical parameters

C) Another very critical parameter is regularizer.

A) 3D contour visualization



Dark blue → Highest Depth

Red → lowest Depth

Problem of local & Global minima?

In smaller Neural Networks, hinderance of local minima is greater than in the larger neural networks.

As we train larger & deep networks, the problem of local minima becomes significantly less imp. The loss curve gets more smoother.